



Electrothermal Digital Twin and Raman DTS for Solar Inverter Condition Monitoring

Supervisor: Professor Martyn Fice
Tengku Faris Bin Tengku Zuhri

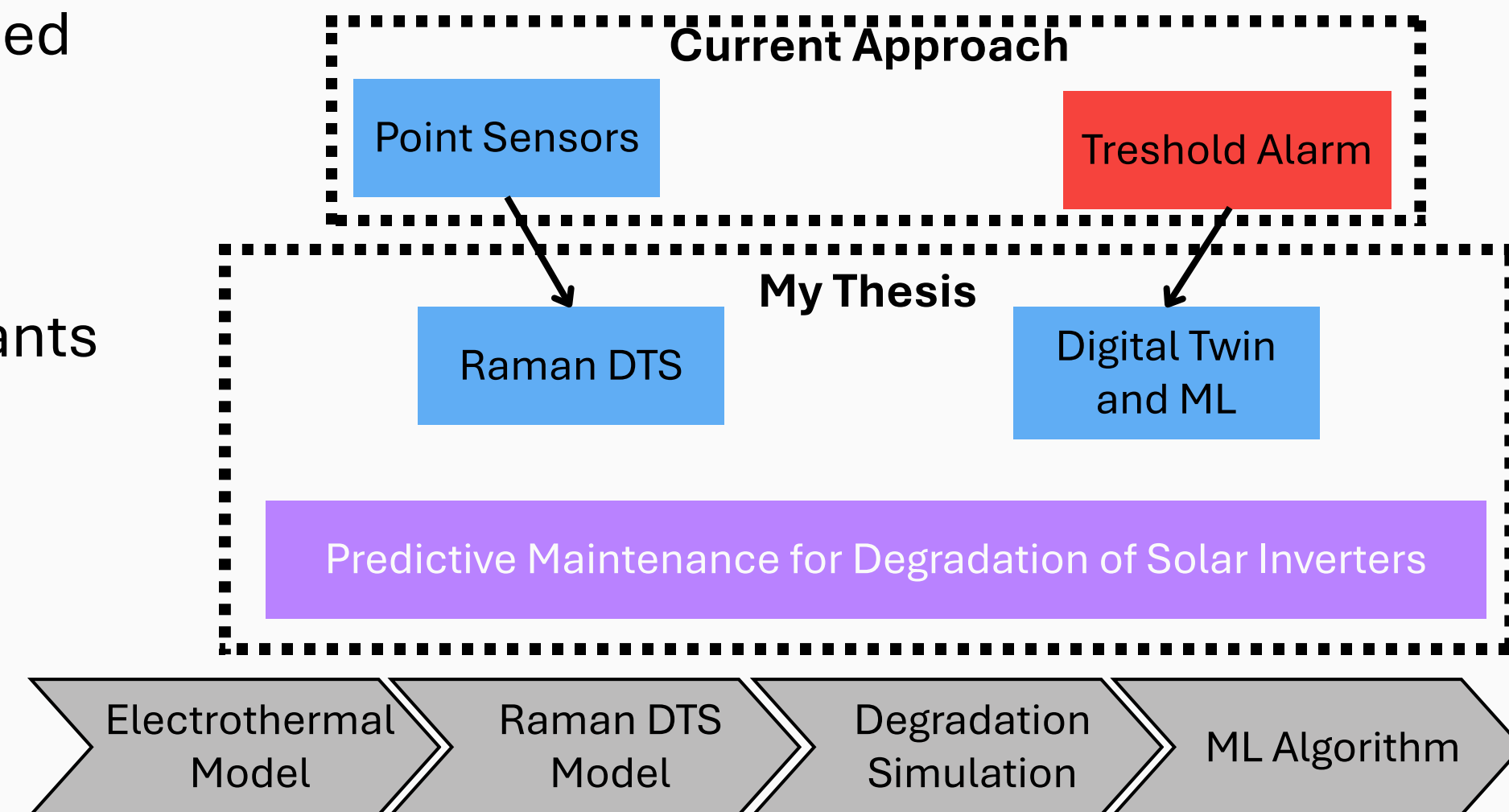


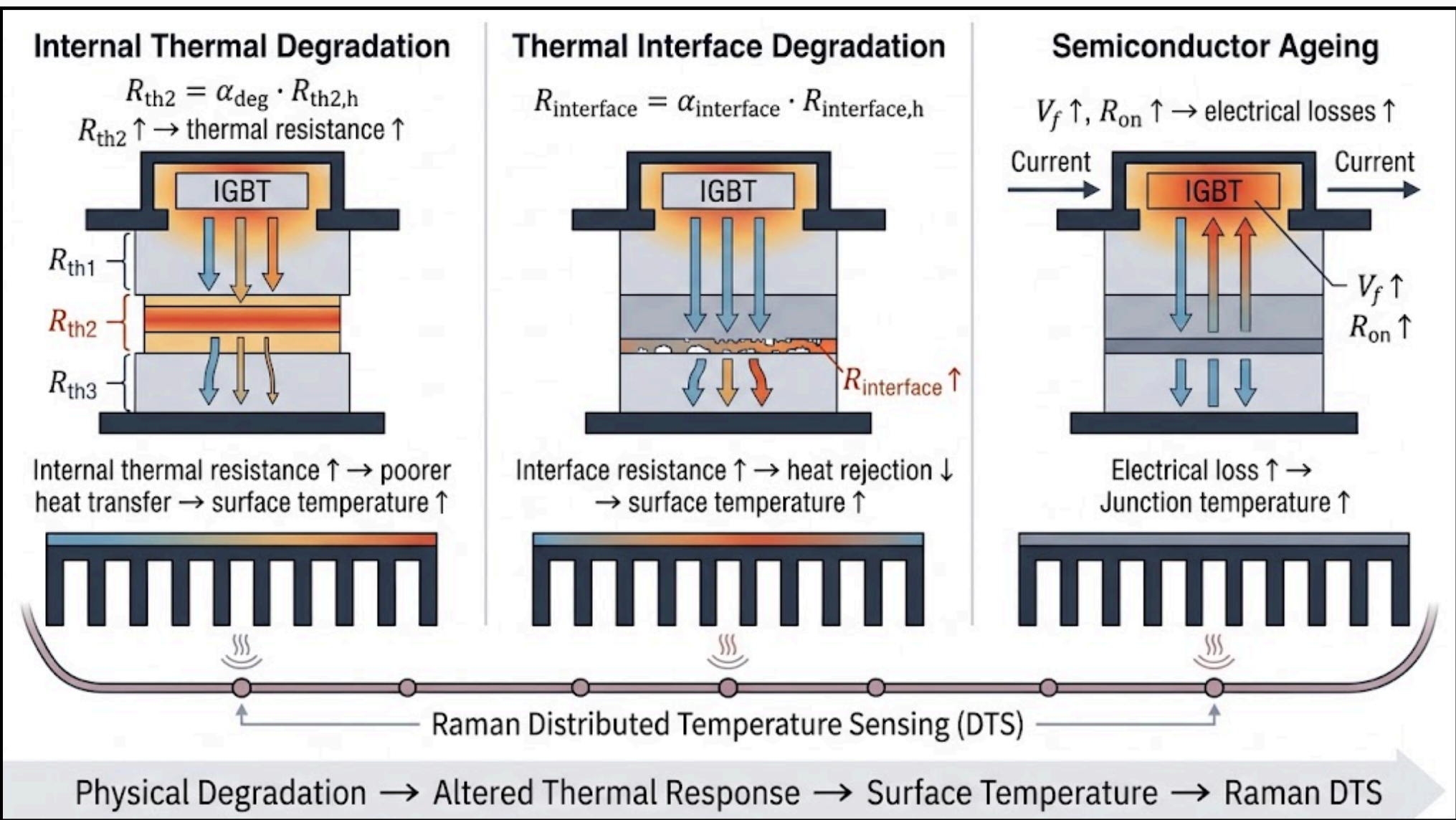
- **Context**

- **Energy Contribution:** Largest growth in global energy supply in 2025 3TW (**25%** increase) [1]
- **Inverters:** Most failure prone component in Solar PV [2]
- **Monitoring:** Reactive and not predictive, limited localisation

- **Recent Literature**

- Trend Based Predictive Maintenance in PV Plants [3]
- Distributed optic fiber Sensing for Large Infrastructure [4]





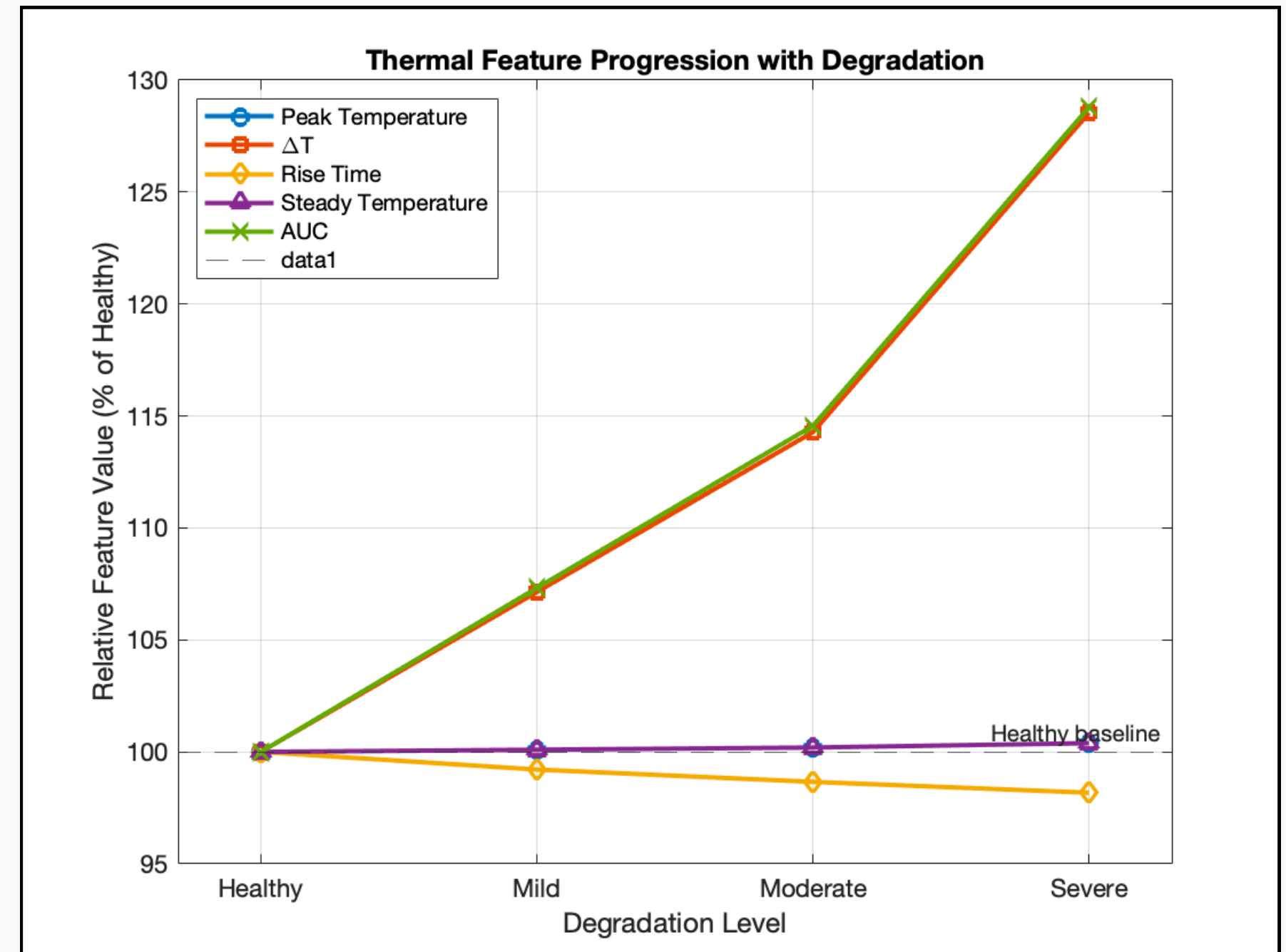
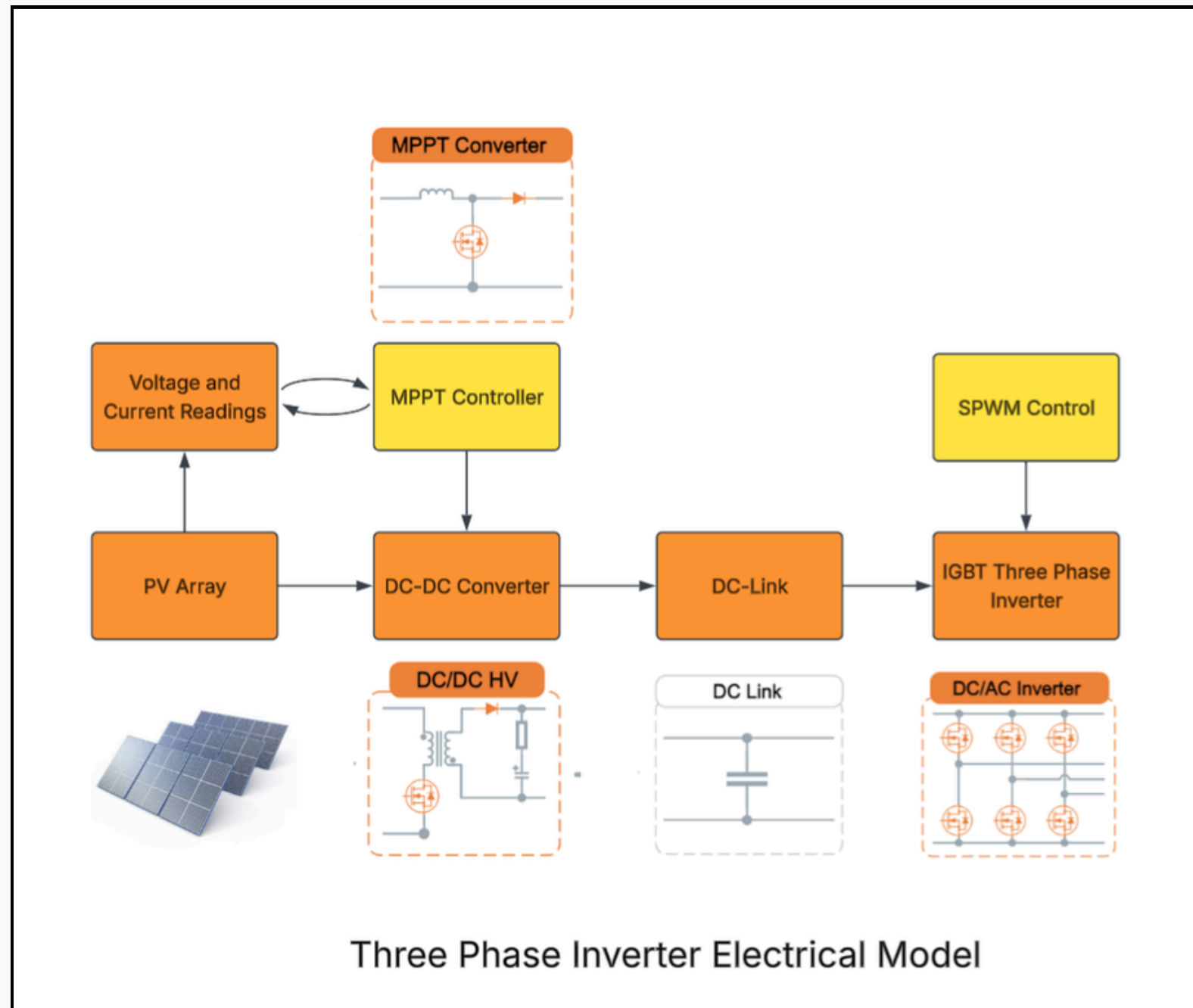
Internal Thermal	Thermal Interface	Semiconductor Ageing
$R_{th2} \uparrow$	$R_{interface} \uparrow$	$V_f, R_{on} \uparrow$
Heat Transfer $\downarrow$	Heat Rejection $\downarrow$	Heat Generation $\uparrow$
+28.8% DTS Rise	DTS Validation Next	DTS Validation Next

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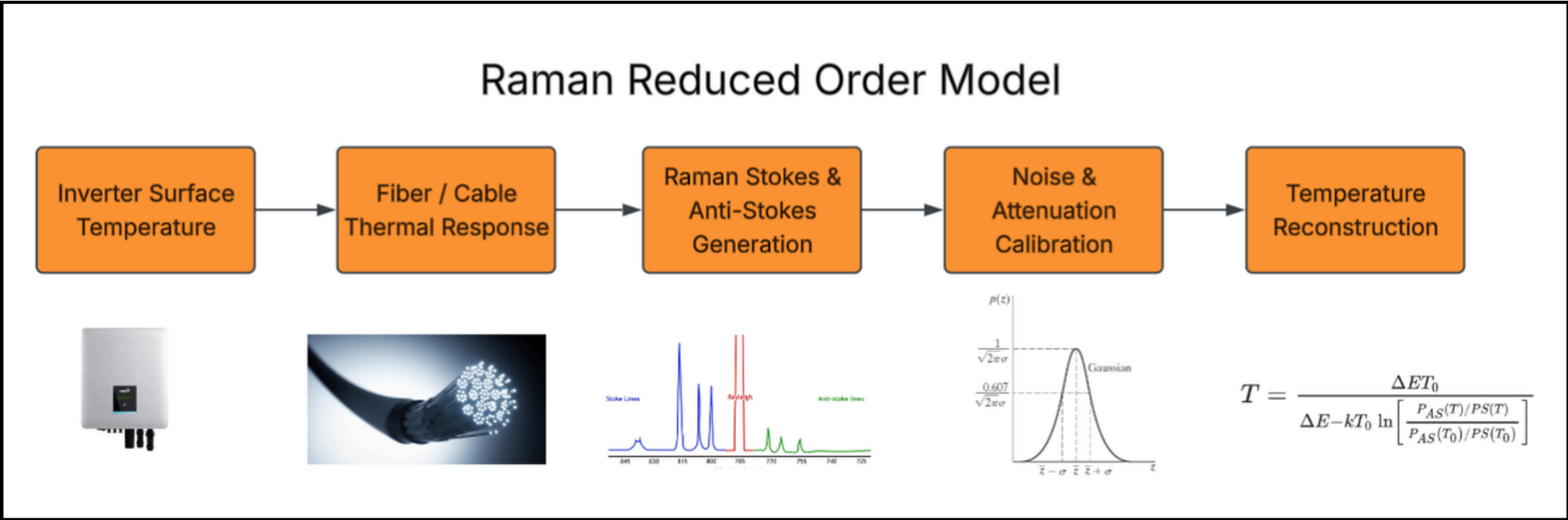
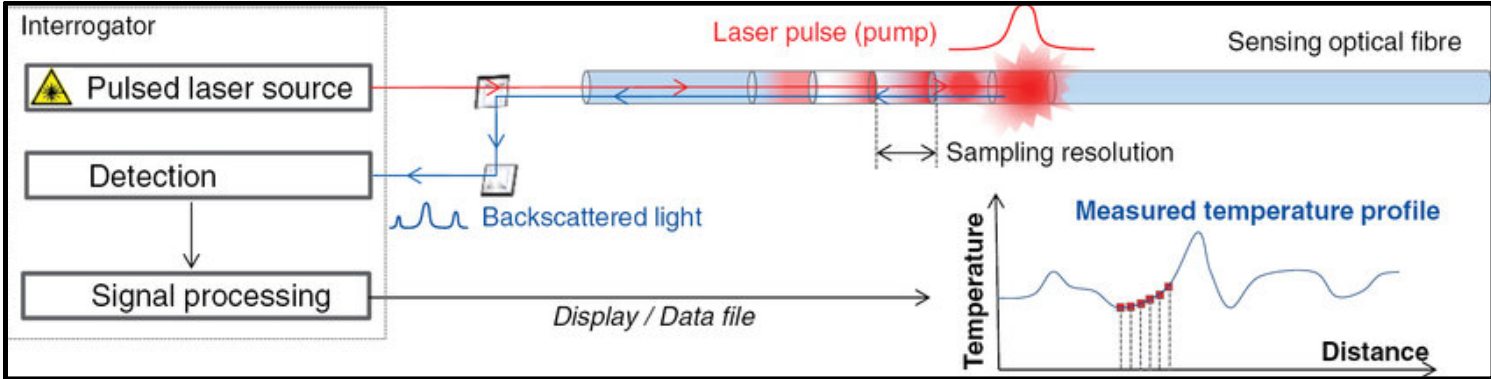
Different forms of inverter degradation create thermal signatures that can **reveal changing health**

Bridges inverter electrical behaviour and thermal dynamics for **realistic degradation modelling**.

Degradation levels produce **distinct** changes in inverter surface



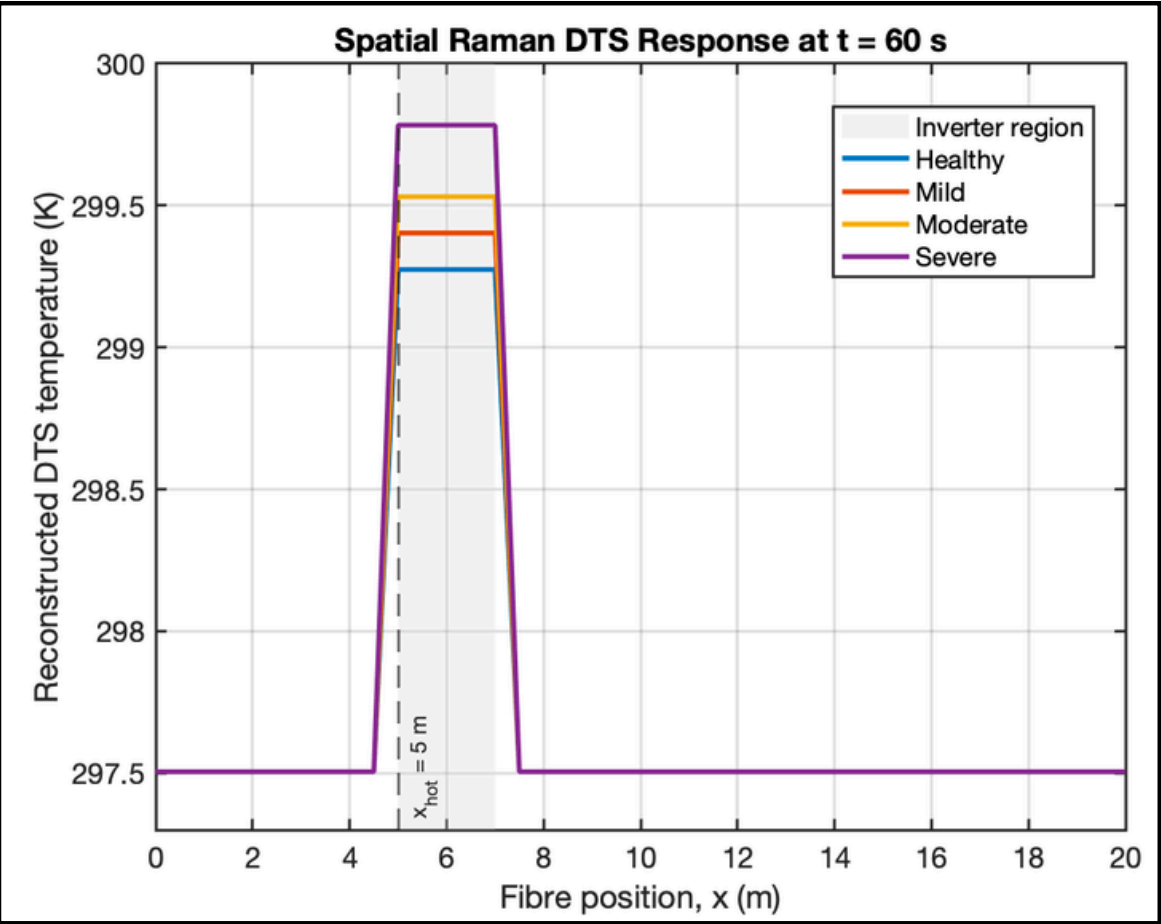
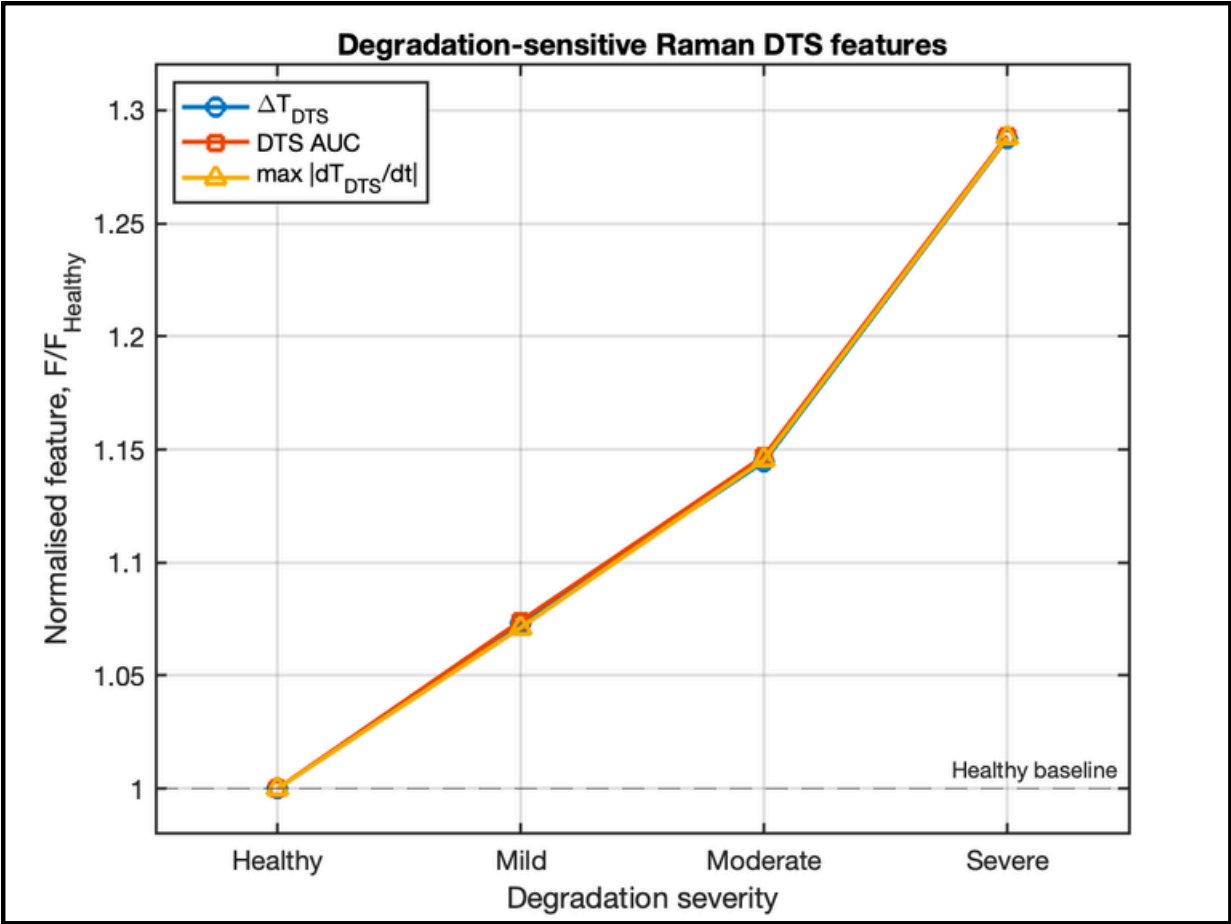
$$\Delta T \uparrow 28.5\% \quad \frac{dT}{dx} \uparrow 28.8\%$$



- Spatial Localisation
- Distinct Trend

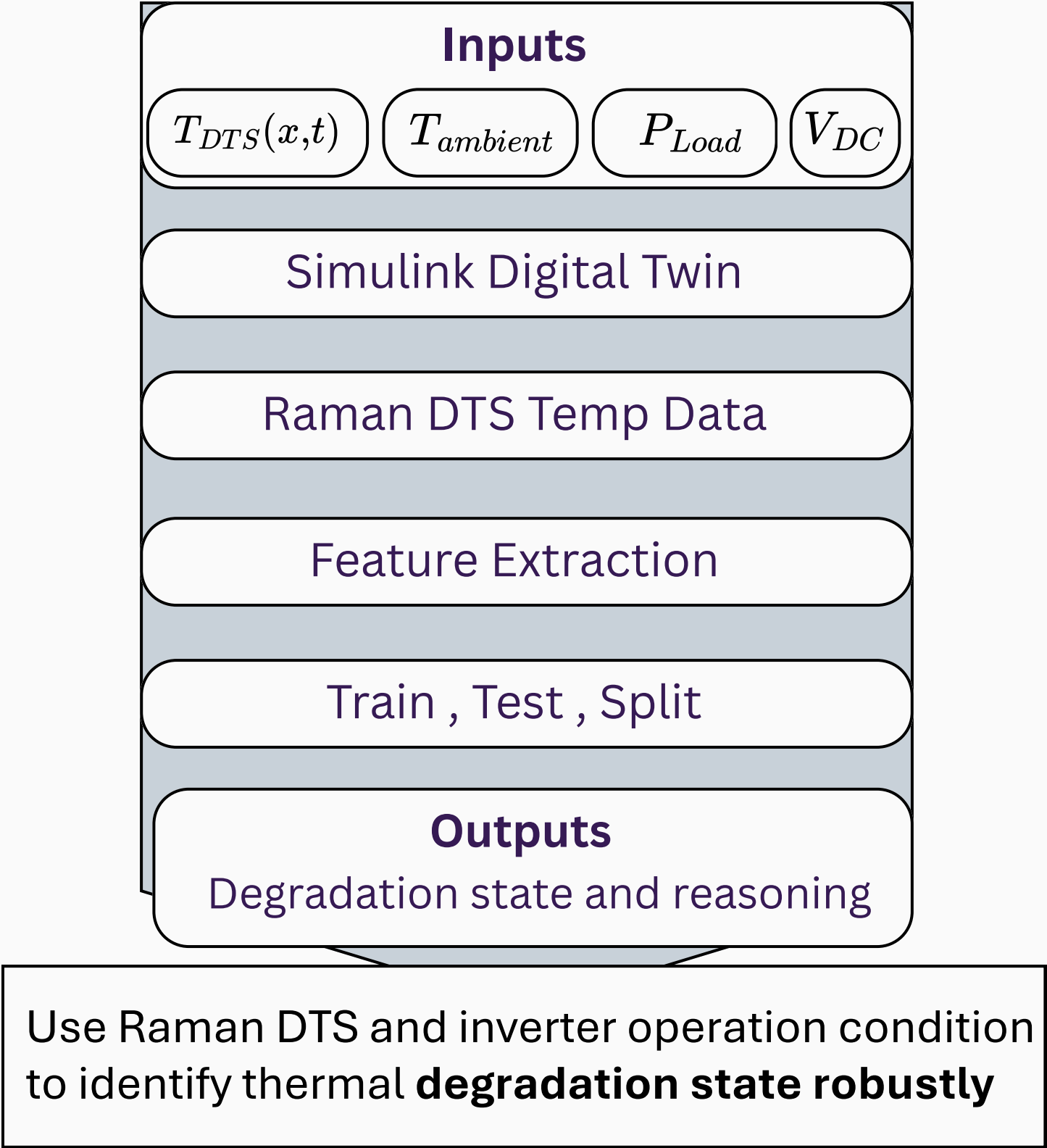
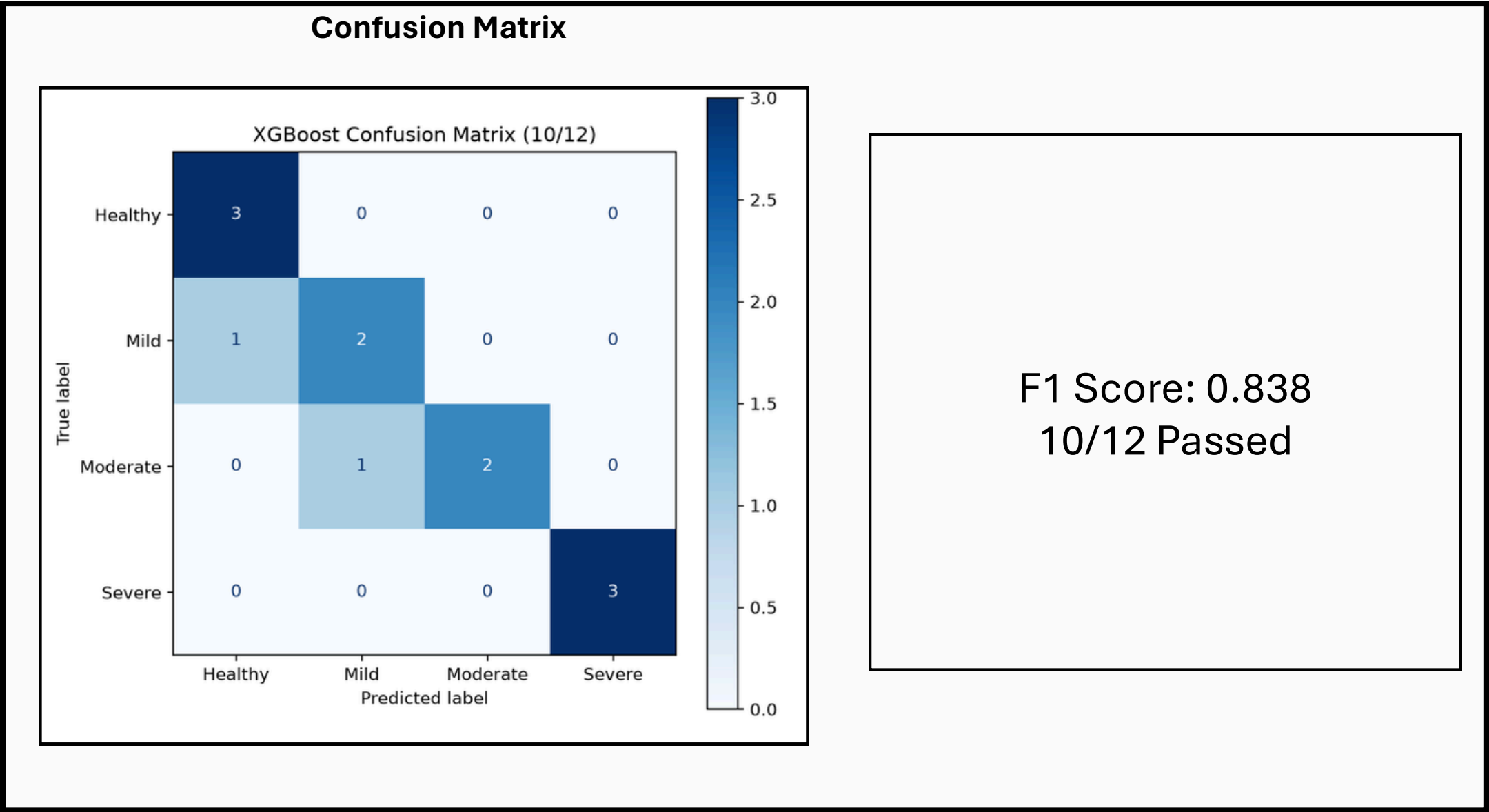
$\Delta T \uparrow 28.8\%$
from healthy to severe

$\frac{dT}{dx} \uparrow 28.9\%$
from healthy to severe



$$\hat{y}_i = \sum_{k=1}^K f_k(\mathbf{x}_i)$$

Sequential decision trees learn nonlinear relationships between DTS thermal signatures, operating conditions and degradation severity.



Research Contributions:

- **Physics-informed electrothermal digital twin**

Simulates four levels of PV inverter thermal-path degradation, with surface temperature rise increasing from **3.509 K to 4.679 K (~33.3%)** from Healthy to Severe.

- **Integrated spatial Raman DTS sensing chain**

Transforms inverter thermal behaviour into distributed fibre measurements across 41 spatial channels, while retaining the degradation trend with a **~28.8% increase** in DTS temperature rise from Healthy to Severe.

- **Operating-condition-aware predictive-maintenance framework**

Moves beyond fixed temperature thresholds by testing degradation across 50 independent simulations spanning 5 ambient temperatures × 5 operating points × 4 degradation severities.

Next Steps:

- **Refined Raman DTS Model (Receiver End) and Extended Length**

Develop a more physically representative sensing model, incorporating realistic signal acquisition, calibrated noise and distance-dependent effects to model non-linear effects more clearly (**Currently in progress**)

- **Train XGBoost Model with Different Degradation Types and Validate with Real World Data**

Generate larger independent simulation datasets for additional thermal degradation mechanisms, such as internal thermal-path and thermal-interface degradation, while varying ambient and operating conditions (**Currently in Progress**)